Bangladesh University of Business and Technology

LAB REPORT

Experiment No : 01
Course Title : Algorithms Lab
Course Code : CSE 232
Experiment Name: Lab Assignment 01

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1. Even or Odd

EvenOdd_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    if (n % 2 == 0)
    {
        cout << n << " is Even";
    }
    else
    {
        cout << n << " is Odd";
    }

    return 0;
}
```

C:\WINDOWS\system32

```
15
15 is Odd
c:\Users\Kawsar Hoque\Desktop\CODING KH\BUBT_LAB
_ALGORITHM\LAB_TASK1>
```

2. Find Maximum of Two Numbers

MaximumTwoNumbers_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a, b;
    cin >> a >> b;

    if (a > b)
    {
        cout << a;
    }
    else
    {
        cout << b;
    }

    return 0;
}
```

C:\WINDOWS\system32\

```
42 89
89
c:\Users\Kawsar Hoque\Desktop
\CODING KH\BUBT_LAB_ALGORITHM
\LAB_TASK1>
```

3. Sum of First N Natural Numbers

SumNaturalNumbers_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    int sum = 0;

    for (int i = 1; i <= n; i++)
    {
        sum += i;
    }

    cout << sum;

    return 0;
}
```

C:\WINDOWS\system32\

```
10
55
c:\Users\Kawsar Hoque\Desktop
\CODING KH\BUBT_LAB_ALGORITHM
\LAB_TASK1>
```

4. Check if a Number is Prime

PrimeCheck_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    bool prime = true;

    if (n <= 1)
    {
        prime = false;
    }
    else
    {
        for (int i = 2; i < n; i++)
        {
            if (n % i == 0)
            {
                prime = false;
                break;
            }
        }
    }

    if (prime)
    {
        cout << "Prime";
    }
    else
    {
        cout << "Not Prime";
    }

    return 0;
}
```

C:\WINDOWS\system32

```
17
Prime
c:\Users\Kawsar Hoque\Desktop\CODING KH\BUBT_LAB
_ALGORITHM\LAB_TASK1>
```

5. Factorial of a Number

Factorial_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    long long fact = 1;

    for (int i = 1; i <= n; i++)
    {
        fact *= i;
    }

    cout << fact;

    return 0;
}
```

C:\WINDOWS\system32\

```
6
720
c:\Users\Kawsar Hoque\Desktop\CODING KH\
BUBT_LAB_ALGORITHM\LAB_TASK1>
```

6. Reverse a Number

ReverseNumber_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    int reverse = 0;

    while (n > 0)
    {
        int digit = n % 10;
        reverse = reverse * 10 + digit;
        n /= 10;
    }

    cout << reverse;

    return 0;
}
```

C:\WINDOWS\system32\cmd.

```
9876
6789
c:\Users\Kawsar Hoque\Desktop\CODING KH
\BUBT_LAB_ALGORITHM\LAB_TASK1>
```

7. Check if Number is Palindrome

PalindromeNumber_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    int original = n;
    int reverse = 0;

    while (n > 0)
    {
        int digit = n % 10;
        reverse = reverse * 10 + digit;
        n /= 10;
    }

    if (original == reverse)
    {
        cout << "Palindrome";
    }
    else
    {
        cout << "Not Palindrome";
    }

    return 0;
}
```

C:\WINDOWS\system32

```
1331
Palindrome
c:\Users\Kawsar Hoque\Desktop\CODING KH
\BUBT_LAB_ALGORITHM\LAB_TASK1>
```

8. Print First N Fibonacci Numbers

FibonacciSeries_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    int a = 0, b = 1;

    for (int i = 1; i <= n; i++)
    {
        cout << a << " ";

        int next = a + b;
        a = b;
        b = next;
    }

    return 0;
}
```

C:\WINDOWS\system32\cmd.

```
7
0 1 1 2 3 5 8
c:\Users\Kawsar Hoque\Desktop\CODING KH\BUBT_LAB
_ALGORITHM\LAB_TASK1>
```

9. Count Digits of a Number

CountDigits_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    int count = 0;

    while (n > 0)
    {
        count++;
        n /= 10;
    }

    cout << count << " digits";

    return 0;
}
```

C:\WINDOWS\system32

```
2024
4 digits
c:\Users\Kawsar Hoque\Desktop\CODING KH\BUBT_LAB
_ALGORITHM\LAB_TASK1>
```

10. Sum of Digits of a Number

SumOfDigits_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    int sum = 0;

    while (n > 0)
    {
        sum += n % 10;
        n /= 10;
    }

    cout << sum;

    return 0;
}
```

C:\WINDOWS\system32

```
512
8
c:\Users\Kawsar Hoque\Desktop\CODING KH\BUBT_LAB
_ALGORITHM\LAB_TASK1>
```

11. Linear Search

LinearSearch_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    int arr[n];

    for (int i = 0; i < n; i++)
    {
        cin >> arr[i];
    }

    int key;
    cin >> key;

    bool found = false;

    for (int i = 0; i < n; i++)
    {
        if (arr[i] == key)
        {
            cout << "Found at index " << i;
            found = true;
            break;
        }
    }

    if (!found)
    {
        cout << "Not Found";
    }

    return 0;
}
```

C:\WINDOWS\system32\cmd.

```
4
10 20 30 40
30
Found at index 2
c:\Users\Kawsar Hoque\Desktop\CODING KH\BUBT_LAB_ALGORITHM\LAB_TASK1>
```

12. Swap Two Numbers

SwapNumbers_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a, b;
    cin >> a >> b;

    int temp = a;
    a = b;
    b = temp;

    cout << "a = " << a << endl;
    cout << "b = " << b;

    return 0;
}
```

C:\WINDOWS\system3:

```
10 20
a = 20
b = 10
c:\Users\Kawsar Hoque\Desktop\CODING KH\BUBT_LAB_ALGORITHM\LAB_TASK1>
```

13. Simple Calculator

SimpleCalculator_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a, b;
    char op;

    cin >> a >> op >> b;

    if (op == '+')
    {
        cout << "Result: " << a + b;
    }
    else if (op == '-')
    {
        cout << "Result: " << a - b;
    }
    else if (op == '*')
    {
        cout << "Result: " << a * b;
    }
    else if (op == '/')
    {
        cout << "Result: " << a / b;
    }
    else
    {
        cout << "Invalid Operator";
    }

    return 0;
}
```

C:\WINDOWS\system3:

```
10 * 5
Result: 50
c:\Users\Kawsar Hoque\Desktop\CODING KH
\BUBT_LAB_ALGORITHM\LAB_TASK1>
```

14. Print Even Numbers from 1 to N

PrintEvenNumbers_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    for (int i = 1; i <= n; i++)
    {
        if (i % 2 == 0)
        {
            cout << i << " ";
        }
    }

    return 0;
}
```

C:\WINDOWS\system32

```
10
2 4 6 8 10
c:\Users\Kawsar Hoque\Desktop\CODING KH
\BUBT_LAB_ALGORITHM\LAB_TASK1>
```

15. Find Second Largest Number

SecondLargest_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    int arr[n];

    for (int i = 0; i < n; i++)
    {
        cin >> arr[i];
    }

    int largest = arr[0];
    int secondLargest = -1;

    for (int i = 1; i < n; i++)
    {
        if (arr[i] > largest)
        {
            secondLargest = largest;
            largest = arr[i];
        }
        else if (arr[i] > secondLargest && arr[i] != largest)
        {
            secondLargest = arr[i];
        }
    }

    cout << "Second Largest: " << secondLargest;

    return 0;
}
```

C:\WINDOWS\system32\cmd.

```
6
12 45 7 34 45 10
Second Largest: 34
c:\Users\Kawsar Hoque\Desktop\CODING KH\BUBT_LAB_ALGORITHM
\LAB_TASK1>
```

16. Find Average of Array Elements

AverageArray_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    int arr[n];
    int sum = 0;

    for (int i = 0; i < n; i++)
    {
        cin >> arr[i];
        sum += arr[i];
    }

    double average = (double)sum / n;

    cout << "Average: " << average;

    return 0;
}
```

C:\WINDOWS\system32\cmd.

```
4
10 20 30 40
Average: 25
c:\Users\Kawsar Hoque\Desktop\CODING KH\BUBT_LAB_
ALGORITHM\LAB_TASK1>
```

17. Remove Duplicates from Array

RemoveDuplicates_270.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    int arr[n];

    for (int i = 0; i < n; i++)
    {
        cin >> arr[i];
    }

    for (int i = 0; i < n; i++)
    {
        bool duplicate = false;

        for (int j = 0; j < i; j++)
        {
            if (arr[i] == arr[j])
            {
                duplicate = true;
                break;
            }
        }

        if (!duplicate)
        {
            cout << arr[i] << " ";
        }
    }

    return 0;
}
```

C:\WINDOWS\system32\cmd.

```
7
1 2 2 3 4 4 5
1 2 3 4 5
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\BUBT_LAB_ALGORITHM\LAB_TASK1>
```

18. Convert Lowercase to Uppercase

LowercaseToUppercase_270.cpp > main()

```cpp
#include <iostream>
#include <string>
using namespace std;

int main()
{
    string s;
    getline(cin, s);

    for (int i = 0; i < s.size(); i++)
    {
        if (s[i] >= 'a' && s[i] <= 'z')
        {
            s[i] = s[i] - 32;
        }
    }

    cout << s;

    return 0;
}
```

C:\WINDOWS\system32\cmd.

```
hello world
HELLO WORLD
c:\Users\Kawsar Hoque\Desktop\CODING KH\BUBT
_LAB_ALGORITHM\LAB_TASK1>
```